

Power required BEM

Forward flight

Rotor profile drag + rotor drag power

Rotor profile drag power is the power required to overcome the blades aerodynamic drag force due to friction

$$P_p = \frac{N}{2\pi} \int_0^{2\pi} d\psi \int_0^R dD_p \cdot \Omega r$$

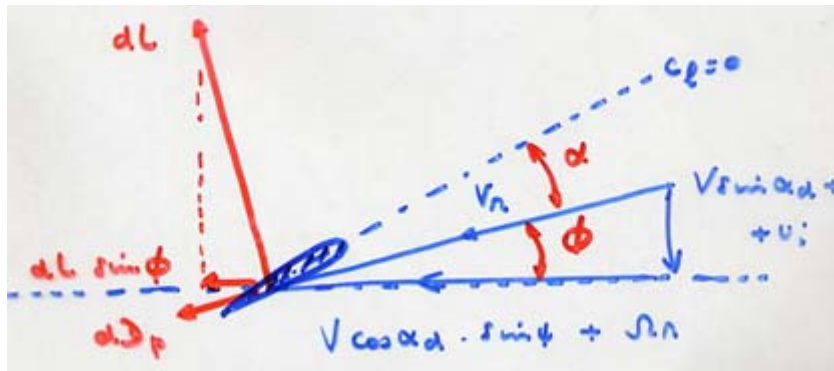
Rotor drag power = The power required to overcome the difference in profile drag forces between the advancing and retreating side (the H_0 force)

$$P_d = H_0 \cdot V \cos \alpha_d$$

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$$P_p = \frac{N}{2\pi} \int_0^{2\pi} d\psi \int_0^R dD_p \cdot \Omega r \quad (1)$$



$$dD_p = \frac{\rho}{2} V^2 C_{Dp} c dr = \quad (2)$$

$$\frac{\rho}{2} (\Omega r + V \cos \alpha_d \sin \psi)^2 C_{Dp} c dr$$

Substituting (2) into (1):

Rotor profile drag power

$$P_p = Nc \frac{1}{2\pi} \bar{C}_{D_p} \int_0^{2\pi} d\psi \int_0^R \frac{\rho}{2} dr (\Omega r + V \cos \alpha_d \sin \psi)^2 \cdot \Omega r \quad (3)$$

This integral can be solved. How?

Advance ratio

$$\mu = \frac{V \cos \alpha_d}{\Omega R} \cong \frac{V}{\Omega R} = 0 \dots 0.35 \quad (4)$$

Rotor profile drag power

$$P_p = Nc \frac{1}{2\pi} \bar{C}_{D_p} \int_0^{2\pi} d\psi \int_0^R \frac{\rho}{2} dr (\Omega r + V \cos \alpha_d \sin \psi)^2 \cdot \Omega r \quad (3)$$

Non-dimensionalizing $x = \frac{r}{R} \quad \mu = \frac{V \cos \alpha_d}{\Omega R} \cong \frac{V}{\Omega R}$

and recalling: $\sigma = \frac{Nc}{\pi R}$

$$P_p = \frac{\sigma \bar{C}_{D_p}}{2} \rho (\Omega R)^3 \pi R^2 \cdot \frac{1}{2\pi} \int_0^{2\pi} d\psi \int_0^1 (x + \mu \sin \psi)^2 x \cdot dx \quad (5)$$

Solving integral (5) finally results in:

Rotor profile drag power

$$P_p = \frac{\sigma \bar{C}_{Dp}}{8} \rho (\Omega R)^3 \pi R^2 (1 + \mu^2) = P_{phov} \cdot (1 + \mu^2) \quad (6)$$

Rotor drag power

Rotor drag power = The power required to overcome the difference in profile drag forces between the advancing and retreating side (the H_0 force)

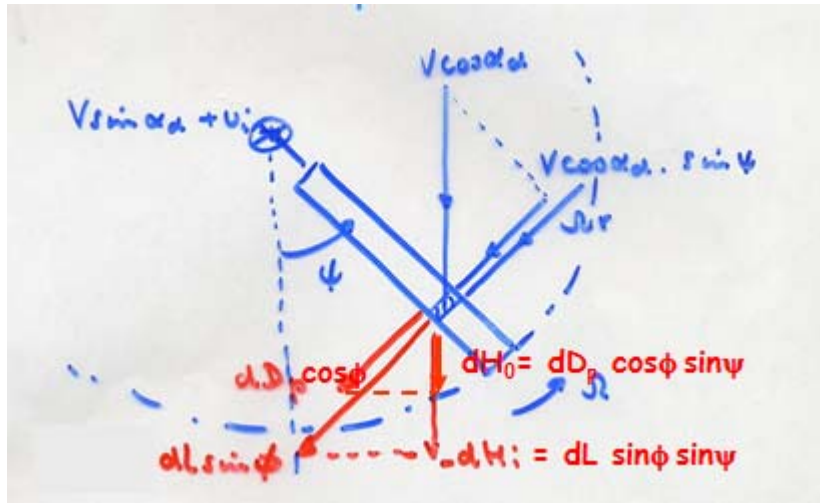
$$P_d = H_0 \cdot V \cos \alpha_d \quad (7)$$

Using the definition of advance ratio $\mu = \frac{V \cos \alpha_d}{\Omega R} \cong \frac{V}{\Omega R}$ one obtains:

$$P_d = H_0 \cdot (\Omega R) \mu \quad (8)$$

We need to calculate the H_0 force first.

Rotor drag power - H0 force calculation



$$H_0 = \frac{N}{2\pi} \int_0^{2\pi} d\psi \int_0^R dD_p \cdot \sin \psi \cos \phi \quad (9)$$

Recall again:

$$dD_p = \frac{\rho}{2} cdr (\Omega r + V \cos \alpha_d \sin \psi)^2 C_{Dp} \quad (10)$$

Substituting (10) into (9) and integrating (9) gives H_0 force as:

$$H_0 = \frac{\sigma \bar{C}_{Dp}}{4} \rho (\Omega R)^2 \pi R^2 \cdot \mu \quad (11)$$

Rotor drag power

Knowing the expression of H_0 force (11) and substituting this into the expression of rotor drag power (8) gives:

$$P_d = \frac{\sigma \bar{C}_{Dp}}{4} \rho (\Omega R)^3 \pi R^2 \cdot \mu^2 \quad (12)$$

Rotor Profile drag + Drag power

Usually the expression of rotor profile drag power is combined with the expression of rotor drag power. Adding (6) and (12) gives:

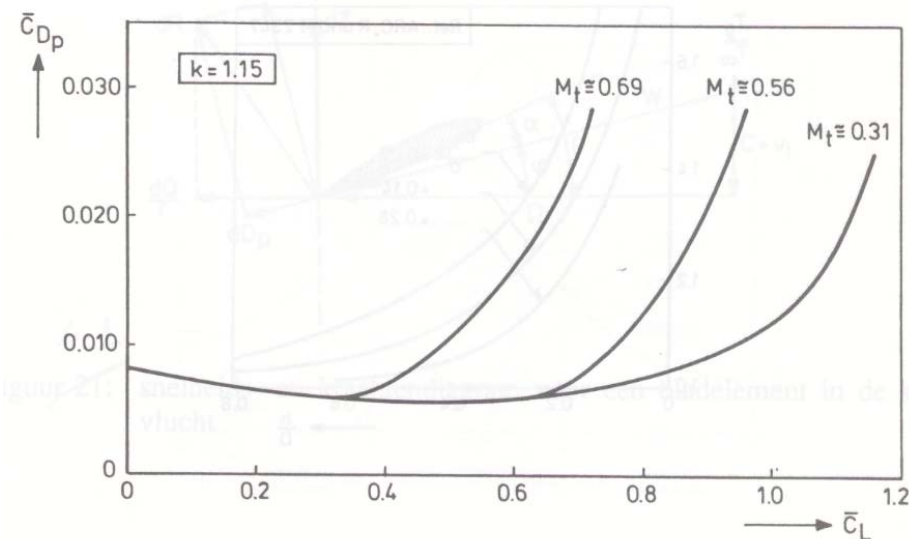
$$P_p + P_d = \underbrace{\frac{\sigma C_{Dp}}{8} \rho (\Omega R)^3 \pi R^2}_{P_{phov}} \cdot (1 + 3\mu^2) \quad (13)$$

Usually this expression is underestimated. Bennett proposed experimentally to use a correction factor of 4.65 in the so-called Bennett approximation:

$$P_p + P_d = P_{phov} (1 + 4.65\mu^2) \quad (14)$$

How to determine blade medium drag coefficient?

$\bar{C}_{Dp}$ can be determined graphically as a function of medium lift coefficient



Recall that $\bar{C}_L = \frac{6.6C_T}{\sigma}$ where the thrust coefficient can be approximated

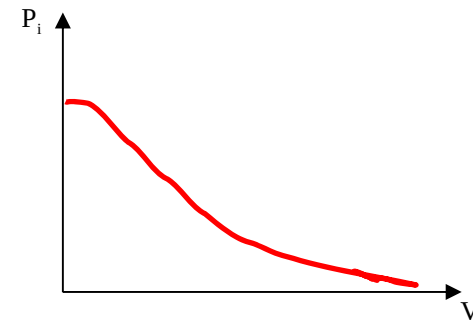
as the weight coefficient $C_T = C_W = \frac{W}{\rho\pi R^2(\Omega R)^2}$

Induced power

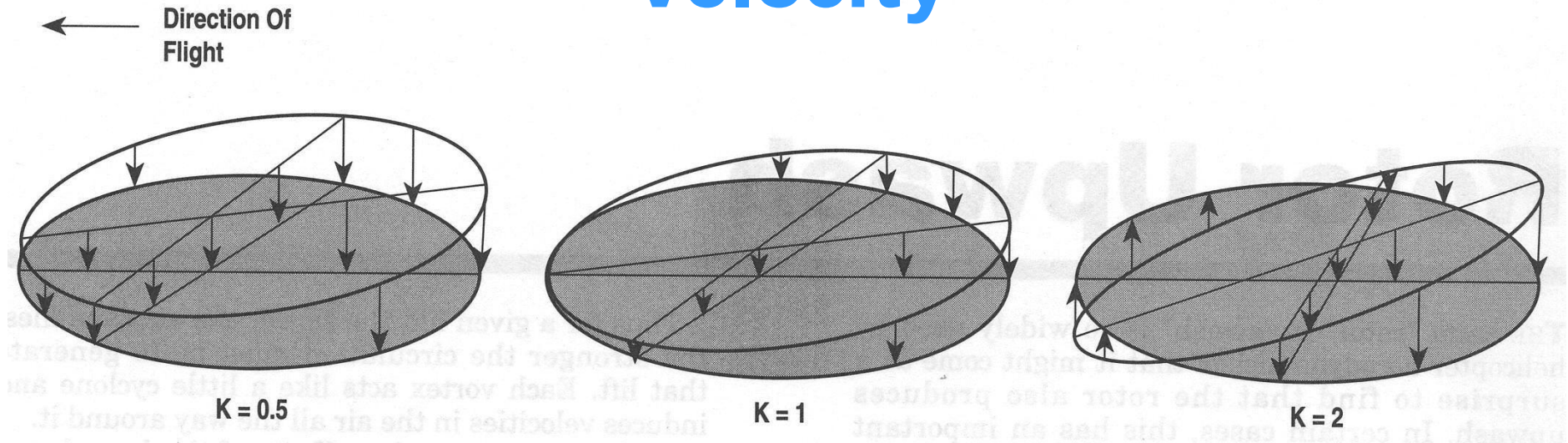
Induced power is the power required to overcome the induced drag force

$$P_i = \frac{N}{2\pi} \int_0^{2\pi} d\psi \int_0^R v_i dT = kP_{id} = kTv_i \quad (15)$$

- Induced power is the power required to move a mass of air through the disk at the induced velocity.
- Induced power decreases with the speed V of the mass flow
- As helicopter goes faster, more air goes through the stream tube and the rotor has to do less work (induced power decreases). At high speeds induced power is $\frac{1}{4}$ of the induced power in hover.



Glauert distribution for induced velocity



- ❑ At the centre of the rotor the induced velocity equals that of the momentum theory (related to the circulation of an airplane wing). The slower the helicopter flies the higher this quantity is
- ❑ From the centre, the induced velocity is sloped linearly (slope constant K) being less at the leading edge of the disc and more at the rear

Fuselage parasite drag power

Parasite power = the power required to overcome the drag of all aircraft except the rotor blades

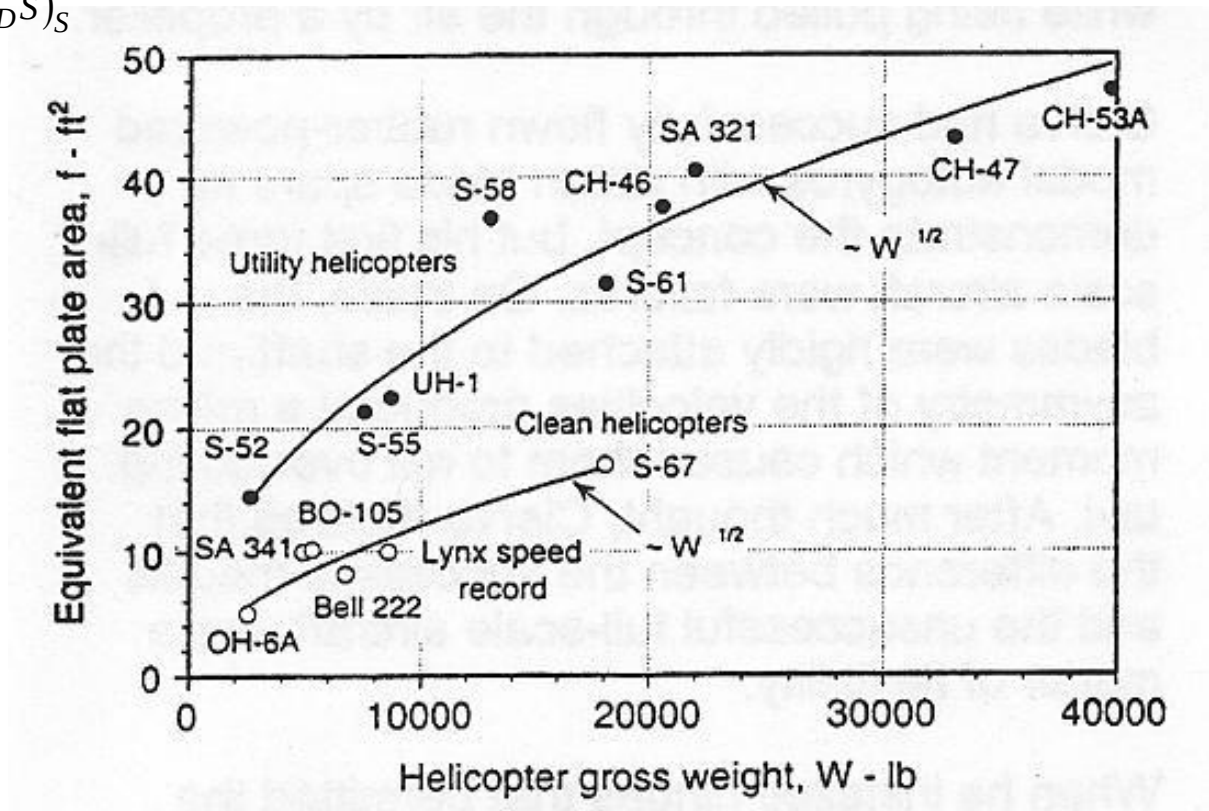
$$P_{par} = D_{par}V = \sum (C_D S)_S \frac{1}{2} \rho V^3$$

- This power is, exactly as in the case of fixed-wing aircraft, depending on the equivalent flat plate drag area $A_{eq} = \sum (C_D S)_S$ (also called parasite drag area)
- At low speed parasite power is not important (only a small fuselage drag force exists as the fuselage is immersed in the wake). At high speeds, parasite drag power is important as it increases as V^3

Equivalent flat plate area

Equivalent flat plate area = a fictitious body having $A_{\text{frontal}}=1$ ft² and a drag coeff. $C_d=1$

$$\sum(C_D S)_s$$



Parasite drag area at helicopters

<i>Helicopter</i>	<i>OH6A</i>	<i>UH-1B</i>	<i>CH-47</i>
Design gross weight	2,550	9,500	33,000
Main rotor disc area	550	1,520	5,900
<i>Components</i>			
Fuselage and engine nacelles	1.5	5.0	16.1
Rotor hubs	1.2	5.5	14.1
Landing gear	0.5	3.0	7.9
Empennage	0.1	0.9	—
Miscellaneous	<u>1.7</u>	<u>5.1</u>	<u>5.1</u>
TOTAL	5.0	19.5	43.2

Helicopters $A_{eq}=5\text{ft}^2$ to 100ft^2

Airplanes $A_{eq}=1\text{ft}^2$ to 10ft^2

Forward flight Power BEM

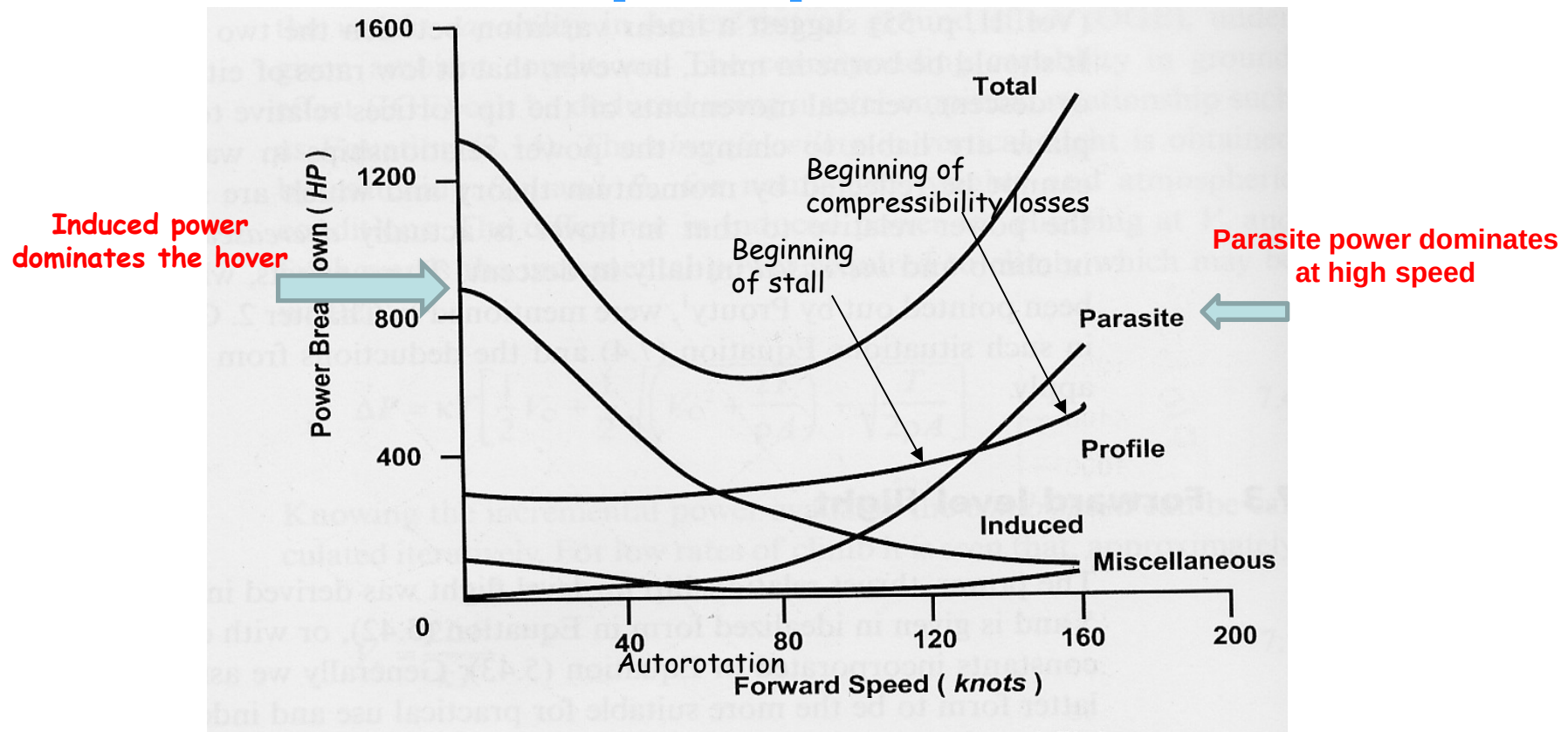
Total power required in forward flight (BEM)

$$P = P_i + P_{\text{par}} + P_p + P_d + P_{\text{misc}}$$

induced power Rotor Parasite power Rotor Profile-drag power Fuselage Drag power Miscellaneous
 (Tail rotor, gearboxes, hydraulic pumps, electrical generators)

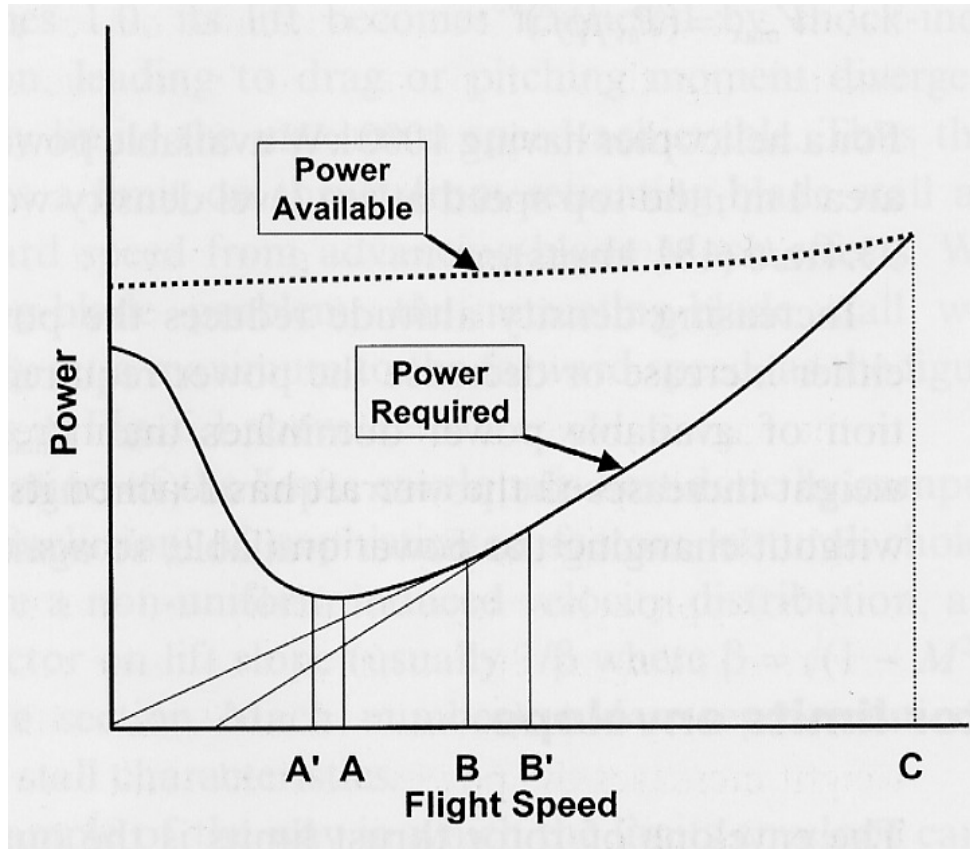
$$P = kT v_i + \frac{\sigma \bar{C}_{Dp}}{8} \rho (\Omega R)^3 \pi R^2 (1 + 4.65 \mu^2) + \sum (C_D S)_s \frac{1}{2} \rho V^3 + P_{\text{misc}}$$

Helicopter power curve



One can see a typical “**bucket**” shape: high in hover falling to a minimum at a moderate speed and rising rapidly at high speeds

Best speeds



Point A defined by the minimum power:

- speed for best endurance (loiter time)
- minimum rate of descent in autorotation
- maximum rate of climb (correction A' if the parasite drag is increased appreciably in climbing)

Point B defined by the tangent to the curve:

- speed for best (maximum) range
- in a head-wind point B'